FOR A COUNTABLE MODULE, FINITE GENERATION IS EQUIVALENT TO THE CANONICAL MAP M TENSOR R^N -> M^N BEING SURJECTIVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The generator extraction is 'as in the proof of Algebra, Proposition algebra-proposition-fg-tensor' - the same argument applied to the tuple (m_n) instead of an arbitrary element of M^N; content verbatim.

Editorial note. The label lemma-countable-fg-tensor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The canonical map M \otimes_R R^N -> M^N (evaluate a finite tensor sum coordinatewise); Algebra, proposition-fg-tensor: for finitely generated M the map is surjective and its proof extracts generators from one tensor element
- Countable modules: elements can be enumerated m_1, m_2, m_3, ...

Source reference. Original proposition 059J: sources/stacks-chow/tags.txt, source lines 6817–6817. Prerequisite pointer invoked at 971; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let M be an R-module which is countable. Then M is a finite R-module if and only if $M \otimes_R R^{\mathbf{N}} \to M^{\mathbf{N}}$ is surjective.*

Proof. If M is a finite module, then the map is surjective by Algebra, Proposition 059J. Conversely, assume the map is surjective. Let $m_1, m_2, m_3, \ldots$ be an enumeration of the elements of M. Let $\sum_{j=1,\ldots,m} x_j \otimes a_j$ be an element of the tensor product mapping to the element $(m_n) \in M^{\mathbf{N}}$. Then we see that $x_1, \ldots, x_m$ generate M over R as in the proof of Algebra, Proposition 059J. □

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